LEDGER · 2026-08-30 · BIOMEDICAL ENGINEERING / MATERIALS SCIENCE

Conductive Inositol Hexaphosphate-Polyaniline (IP6-PANI) Hydrogels via Ambient Aqueous Polymerization

CLEARED

This concept cleared all seven gates and every voiding sub-test.

Incumbent benchmark	Ag/AgCl wet conductive electrode gel (Ten20 paste)
Measured advantage	Impedance shift after 48 hours continuous wear: Incumbent 2.57x increase vs IP6-PANI <1.10x increase (>2.3x better stability)
Time to first revenue	4 months

Gross margin at parity	94.4%
Startup CapEx	$21,000
Payback from first sale	0.8 months
Capital productivity	34.29x
Regulatory risk	Moderate — qualification costs reported ($45,000 estimated)

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Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

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Assessed 2026-08-30 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts